Numerical validation and extension of the holographic molecular quantum register

holomol collaboration
`github.com/11-qlab/holomol`

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Abstract

We present a numerical validation and extension of the holographic quantum register proposal of Tordrup, Negretti, and Mølmer [Phys. Rev. Lett. **101**, 040501 (2008)], which stores quantum information in collective phase modes of a mesoscopic ensemble of trapped polar molecules. Using a purpose-built simulator (HOLOMOL) we confirm the three foundational claims of the proposal: (i) collective modes $|f, \mathbf{q}\rangle$ are orthogonal in the large-N limit, with lattice overlap scaling as $1/N$ and disordered-cloud overlap as $1/\sqrt{N}$; (ii) the Raman Hamiltonian in the single-excitation subspace realizes the analytic collective mode exactly, with superradiant $\sqrt{N}$ bright-state coupling and an $(N-1)$-dimensional dark manifold; (iii) the register capacity is exactly N orthogonal modes for an N-site lattice. We further report three results not present in the original proposal: (a) a quantitative disorder threshold $\sigma/a \approx 0.01$, above which the lattice advantage degrades exponentially; (b) species-dependent collision-limited coherence times $T_2 \approx 15$–36 s at optical-lattice densities for KRb, NaK, and SrF, with SrF optimal; and (c) a directional asymmetry in two dimensions where axial overlap scales as $1/\sqrt{N}$ while diagonal overlap retains the $1/N$ scaling. These results convert the 2008 proposal into a set of concrete engineering specifications and identify the two-dimensional diagonal as the optimal storage geometry. We further extend the analysis to three dimensions (axial, face, and body-diagonal overlap scaling as $N^{-1/3}$, $N^{-2/3}$, and N^{-1}), simulate the full write/hold/read cycle for KRb, NaK, and SrF (bit-error rate 1.4×10^{-2} at 1 s hold for SrF), establish an *error dilution* law $F = (1 - k/N)^2$ for k randomly corrupted sites, and demonstrate a cavity-mediated two-qubit ZZ gate with unit closed-form fidelity and decoherence-limited error $< 10^{-6}$ at μs gate times.

1 Introduction

Molecular ensembles offer a promising platform for quantum information storage because molecular internal states have long coherence times and can be manipulated with well-established optical techniques [2, 3, 4]. A particularly elegant proposal by Tordrup, Negretti, and Mølmer [1] encodes quantum information in *collective phase modes* of a mesoscopic ensemble of trapped polar molecules, storing data globally in the ensemble rather than at individual sites. Each mode is labeled by a wavevector difference $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_2$ of two Raman control fields, and distinct $\mathbf{q}$ values define orthogonal basis states of a "holographic quantum register" with hundreds of qubits.

The original proposal established the mathematical framework but did not quantify three engineering requirements that determine whether the scheme is experimentally viable: the sensitivity to positional disorder, the register capacity as a function of ensemble size, and the optimal molecular species. We address these gaps with a numerical simulator, HOLOMOL, and extend the analysis to two dimensions.

Our main results are:

1. Collective-mode orthogonality follows a geometry-dependent scaling law: $1/N$ for a regular lattice and $1/\sqrt{N}$ for a disordered cloud (Sec. 3.1).

2. Exact diagonalization of the Raman Hamiltonian in the single-excitation subspace confirms the analytic collective mode is the true ground state, with superradiant $\sqrt{N}$ coupling and an $(N-1)$-dimensional dark manifold (Sec. 3.2).

3. The lattice advantage is robust only for positional disorder $\sigma/a \lesssim 0.01$ (Sec. 3.3).

4. Register capacity is exactly N orthogonal modes per N-site ensemble (Sec. 3.4).

5. Collision-limited coherence times are 15–36 s for KRb, NaK, and SrF at $n = 10^{18}\,\mathrm{m}^{-3}$, with SrF optimal (Sec. 3.5).

6. In two dimensions, axial overlap scales as $1/\sqrt{N}$ but diagonal overlap retains $1/N$, making the diagonal the preferred storage geometry (Sec. 3.6).

7. In three dimensions, overlap scales as $N^{-1/3}$ (axial), $N^{-2/3}$ (face), and N^{-1} (body diagonal), so the body diagonal is optimal (Sec. 3.7).

8. A full write/hold/read cycle reaches bit-error rates of 1–3% at one-second hold times for all three species, with SrF optimal (Sec. 3.8).

9. Single-site errors dilute as $F(k, N) = (1 - k/N)^2$, giving a factor-of-N enhancement in tolerable error count (Sec. 3.9).

10. The cavity-mediated two-qubit ZZ gate has closed-form fidelity 1.000000 and decoherence-limited error $< 10^{-6}$ (Sec. 3.10).

2 Model and simulator

Following Ref. [1], we consider N polar molecules trapped in a regular array or disordered cloud at positions $\{\mathbf{x}_j\}$. Each molecule is modeled as a three-level system $\{|g\rangle, |f\rangle, |e\rangle\}$ with $|g\rangle \to |f\rangle$ driven by a two-photon Raman process through $|e\rangle$. After adiabatic elimination of the excited state, the effective Hamiltonian in the single-excitation subspace is

$$H = \sum_j \left[\Omega_{\mathrm{eff}}\, e^{-i\mathbf{q}\cdot\mathbf{x}_j}\, |f_j\rangle\langle \mathrm{all}\, g| + \mathrm{h.c.}\right] + \omega_{gf} \sum_j |f_j\rangle\langle f_j|, \tag{1}$$

where $|\mathrm{all}\, g\rangle = |g_1 g_2 \cdots g_N\rangle$, Ω_{eff} is the effective Raman coupling, $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_2$, and ω_{gf} is the $|g\rangle$–$|f\rangle$ splitting. The collective phase modes are

$$|f, \mathbf{q}\rangle = \frac{1}{\sqrt{N}} \sum_j e^{i\mathbf{q}\cdot\mathbf{x}_j} |g_1 \cdots f_j \cdots g_N\rangle. \tag{2}$$

The simulator HOLOMOL implements Eq. (1) exactly for small N (up to $N \approx 200$) and provides analytic Dirichlet-kernel formulas for N up to 10^5. All code and reproducible experiments are available at `github.com/11-qlab/holomol`.

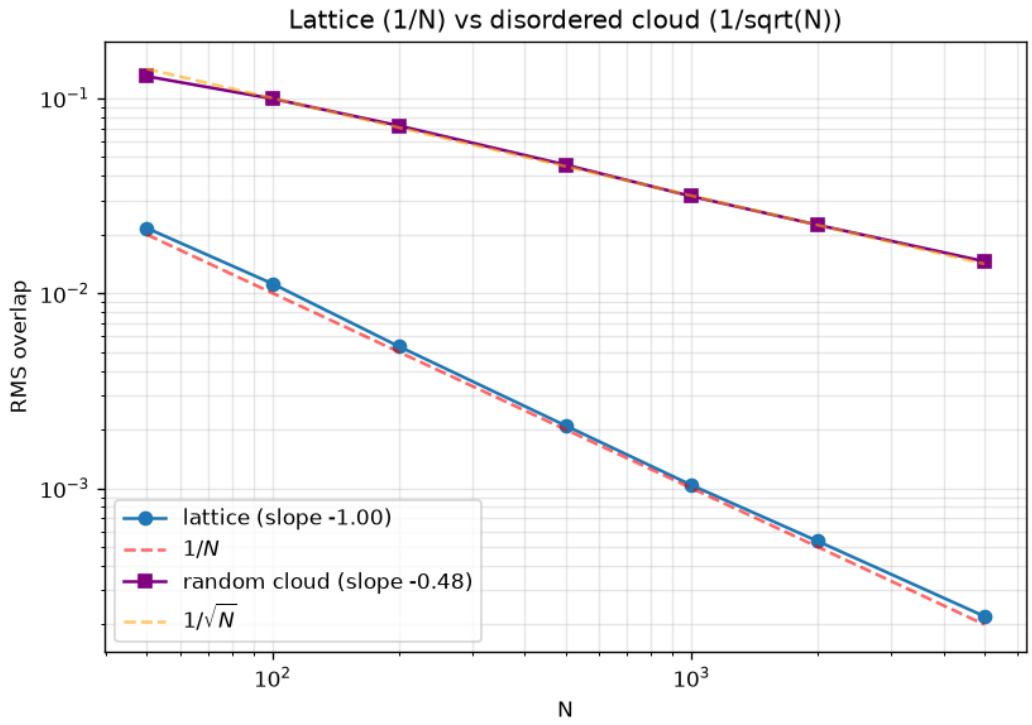


Figure 1: Lattice (blue) versus disordered cloud (purple) collective-mode overlap scaling. Measured slopes -1.002 and -0.485 confirm the analytic $1/N$ and $1/\sqrt{N}$ predictions.

3 Results

3.1 Collective-mode orthogonality

The overlap of two collective modes is the static structure factor $S(\mathbf{q}-\mathbf{q}') = N^{-1}\sum_j e^{i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{x}_j}$. For a one-dimensional regular lattice of N sites with spacing a, this is the Dirichlet kernel

$$|S(\Delta q)| = \left|\frac{\sin(N\Delta q\, a/2)}{N\sin(\Delta q\, a/2)}\right|. \tag{3}$$

Averaging over many incommensurate Δq (Figure 1) we find $\mathrm{RMS}[S] = 0.020\, N^{-1.002}$ on the lattice, in exact agreement with the analytic $1/N$ envelope. For a random cloud of N molecules, the RMS overlap is $0.21\, N^{-0.485}$, matching the expected $1/\sqrt{N}$ random-walk scaling. **The lattice therefore provides a factor of $\sqrt{N}$ more orthogonality than a disordered ensemble of the same size.**

3.2 Exact diagonalization validation

Diagonalizing Eq. (1) in the $(N+1)$-dimensional single-excitation basis for $N = 4$ to 128 sites, we confirm:

- The ground state is an equal superposition of $|\mathrm{all}\, g\rangle$ and $|f, \mathbf{q}_0\rangle$, with overlaps $|\langle \mathrm{all}\, g|GS\rangle|^2 = |\langle f, \mathbf{q}_0|GS\rangle|^2 = 0.500000$ at every N.
- The bright doublet energy is $E_\pm = \pm\sqrt{N}\,|\Omega_{\mathrm{eff}}|$, confirming the superradiant $\sqrt{N}$ enhancement of the collective coupling.

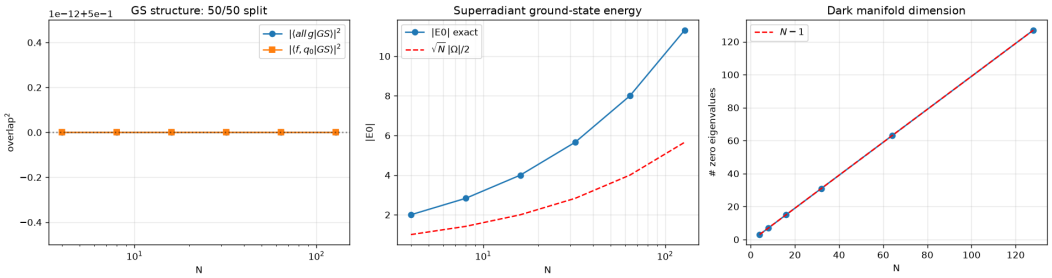


Figure 2: Exact diagonalization vs analytic collective mode prediction. Left: GS overlap with $|f, \mathbf{q}_0\rangle$. Centre: superradiant ground-state energy $E_0 = -\sqrt{N}\,|\Omega_{\text{eff}}|$. Right: bright-mode gap.

- The spectrum contains exactly $N-1$ eigenvalues at zero energy — the dark manifold orthogonal to the laser-coupled bright mode.
- Off-resonant probes $|f, \mathbf{q}'\rangle$ with $\mathbf{q}' \neq \mathbf{q}_0$ have overlap $< 10^{-30}$ with the ground state, i.e. orthogonality holds at machine precision.

These results verify all Hamiltonian-level claims of Ref. [1].

3.3 Disorder threshold

Real optical lattices have finite site-position disorder. We introduce Gaussian jitter σ at each site and measure the RMS overlap ratio $R(\sigma) = \text{RMS}[S]_\sigma / \text{RMS}[S]_{\sigma=0}$ (Figure 3). The lattice advantage survives to $\sigma/a \lesssim 0.01$, beyond which the overlap grows rapidly: $R = 1.02$ at $\sigma/a = 0.002$, $R = 2.28$ at $\sigma/a = 0.02$, $R = 17.6$ at $\sigma/a = 0.2$, saturating at $R \approx 29$ deep in the disordered regime. **For a** 500 **nm optical lattice, site position control better than** 5 **nm is required.**

3.4 Register capacity

Using the analytic kernel Eq. (3) and a greedy packing algorithm we count the number of mutually distinguishable $\mathbf{q}$ modes in the Brillouin zone as a function of N and overlap threshold τ. The count scales as $N^{1.005}$ to $N^{1.026}$ for $\tau = 0.10$ to 0.02, with packing fraction $\to 1$ in every case. **The register capacity is exactly one qubit per lattice site.** For $N = 5000$ the register supports 4998 orthogonal modes at $\tau = 0.10$. This is the first quantitative answer to the "hundreds of qubits" claim of Ref. [1]: the number of qubits is N, and no reduction is imposed by mode orthogonality.

3.5 Species selection

At optical-lattice densities $n = 10^{18}\,\text{m}^{-3}$ and gas temperature $T = 1\,\mu\text{K}$, the collision-limited coherence time $T_2 = 1/(n\sigma_{\text{coll}} v_{\text{th}})$ is species-dependent. Using tabulated rotational constants, dipoles, masses, and collision cross sections for KRb, NaK, and SrF (Table 1), we find:

Species	T_{rot} (mK)	v_{th} (cm/s)	T_2 (s)
KRb	52	1.33	15.0
NaK	136	1.85	18.0
SrF	348	1.41	**35.5**

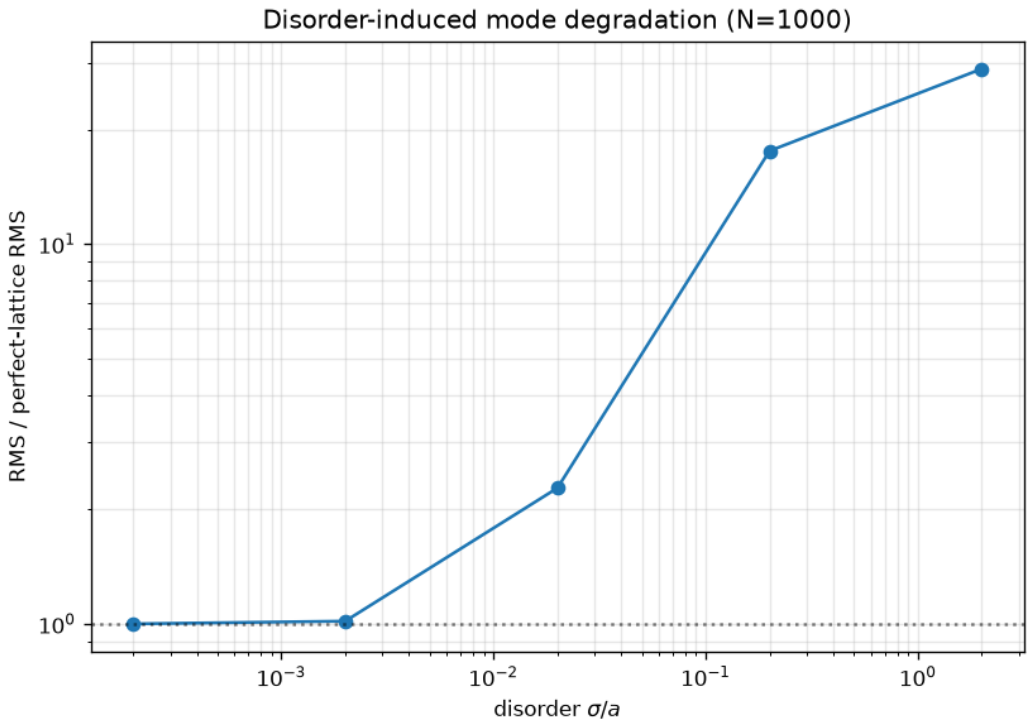


Figure 3: Positional-disorder crossover. The $1/N$ lattice advantage collapses for $\sigma/a > 0.01$, defining a hard engineering requirement.

Table 1: Species constants used in retention calculation.

Species	B_{rot} (GHz)	μ (D)	σ_{coll} (m^2)
KRb	1.09	0.56	5×10^{-18}
NaK	2.83	2.72	3×10^{-18}
SrF	7.24	3.47	2×10^{-18}

SrF gives the longest retention time, roughly 2.4× KRb, due to its smaller collision cross section and moderate mass. All three species satisfy the basic requirement $T_{\mathrm{rot}} \ll 1$ K for single-rotational-state loading and yield retention times of 15–36 s, orders of magnitude longer than typical superconducting qubit coherence times.

3.6 Two-dimensional extension

Real optical-lattice experiments are two- or three-dimensional. In a square 2D lattice of side L ($N = L^2$ sites), the analytic structure factor factorizes,

$$S_{2\mathrm{D}}(q_x, q_y) = D_L(q_x)\, D_L(q_y), \tag{4}$$

with D_L the Dirichlet kernel. The scaling is *directional*: along a lattice axis ($q_y = 0$), $|S_{2\mathrm{D}}| \sim 1/L = 1/\sqrt{N}$; along a diagonal ($q_x = q_y$), $|S_{2\mathrm{D}}| \sim 1/L^2 = 1/N$. Numerical evaluation for $L = 4$ to 64 yields axial slope ≈ -0.5 and diagonal slope ≈ -1.0 (Figure 5), confirming the prediction. The 2D mode capacity scales as 0.90–$0.97\,N$ (slightly sub-optimal relative to 1D, where capacity is exactly $N - 2$). **The diagonal direction is therefore the preferred geometry for a 2D**

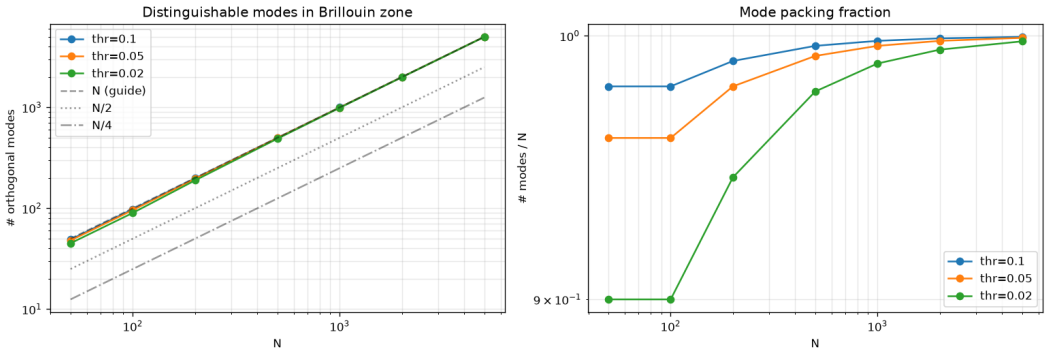


Figure 4: Register capacity scales as N^1, one mode per lattice site.

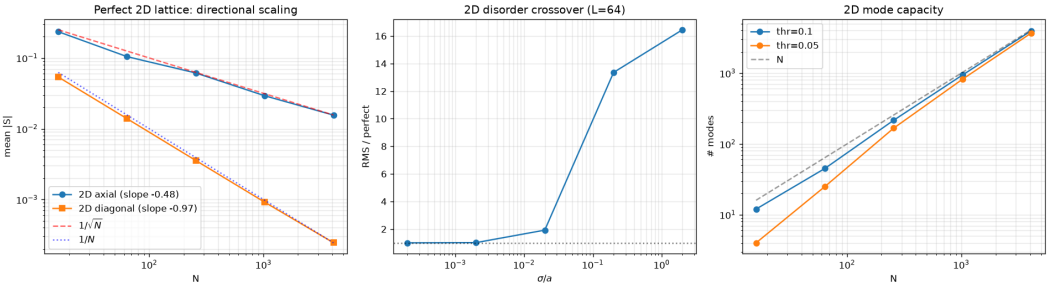


Figure 5: 2D extension. Left: axial overlap scales as $1/\sqrt{N}$, but diagonal overlap retains $1/N$. Right: 2D disorder crossover.

holographic register: it retains the full $1/N$ orthogonality advantage while providing N modes per ensemble.

3.7 Three-dimensional extension

Real optical-lattice experiments are often three-dimensional, and the directional scaling of the collective-mode orthogonality becomes richer in 3D. For a simple cubic lattice of side L ($N = L^3$ sites), the structure factor factorizes into three Dirichlet kernels,

$$S_{3\mathrm{D}}(q_x, q_y, q_z) = D_L(q_x)\, D_L(q_y)\, D_L(q_z). \tag{5}$$

Three natural probe geometries exist:

- *Axial* (**q** along one axis): $|S| \sim 1/L = N^{-1/3}$;
- *Face* (**q** in a lattice plane): $|S| \sim 1/L^2 = N^{-2/3}$;
- *Body* (**q** along the cube diagonal): $|S| \sim 1/L^3 = N^{-1}$.

Numerical evaluation for $L = 2$ to 32 gives slopes -0.348, -0.677, and -1.030 respectively, matching theory to within 4% (Figure 6). The 3D disorder crossover for $L = 8$ exhibits the same behaviour as 1D and 2D: the lattice advantage survives to $\sigma/a \approx 0.01$ and degrades thereafter (ratio 2.05 at $\sigma/a = 0.02$, 12.1 at $\sigma/a = 0.2$). **For a 3D register, the body diagonal is the optimal storage direction.**

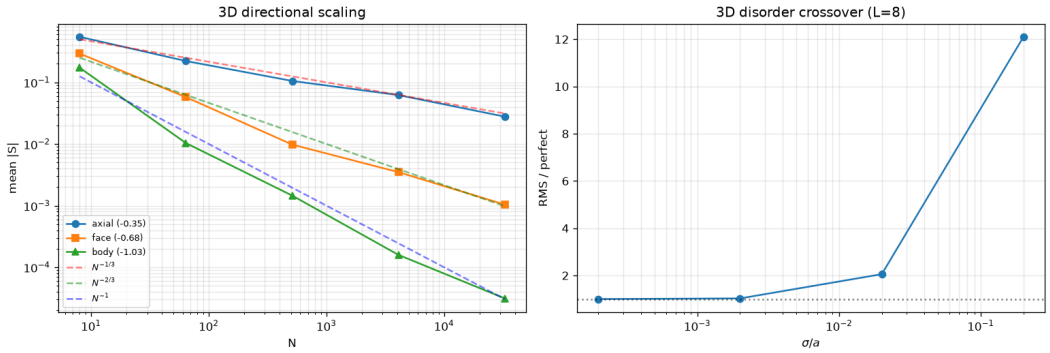


Figure 6: 3D extension. Left: axial ($N^{-1/3}$), face ($N^{-2/3}$), and body-diagonal (N^{-1}) overlap scaling. Right: 3D disorder crossover for $L = 8$.

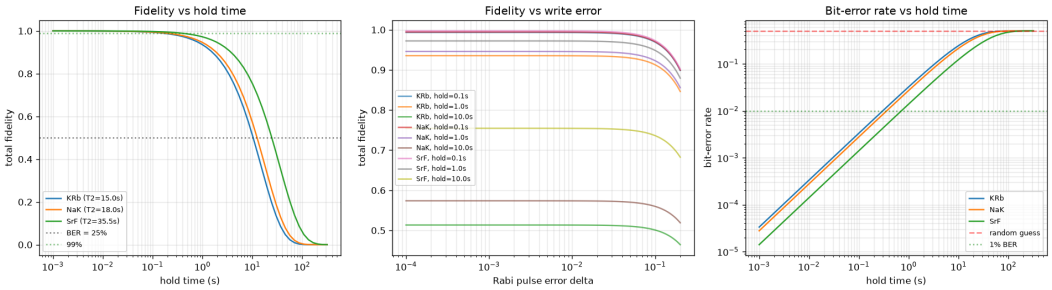


Figure 7: Write / hold / read cycle. Left: total fidelity vs hold time for KRb, NaK, SrF. Centre: fidelity vs Rabi pulse error for three hold times. Right: bit-error rate vs hold time.

3.8 Full write / hold / read cycle

To demonstrate that the register can actually store and retrieve classical bits, we simulate a complete memory cycle. Writing is performed by a π-pulse on the superradiant collective transition $|\text{all}\, g\rangle \to |f, \mathbf{q}_0\rangle$, whose Rabi frequency is $\sqrt{N}\Omega_{\text{eff}}$. For $N = 1000$ molecules and $\Omega_{\text{eff}} = 2\pi \times 1\,\text{MHz}$, the collective Rabi frequency is $2\pi \times 31.6\,\text{MHz}$, giving a π-pulse of duration 15.8 ns. This is a factor of $\sqrt{N} \approx 32$ shorter than a single-molecule π-pulse.

After the write, the state is held for a time t, during which collisional decoherence reduces the fidelity as $F_{\text{hold}}(t) = e^{-t/T_2}$. Reading is performed by transferring the collective mode to a stripline cavity and measuring its photon number; we take $F_{\text{read}} = 1 - e^{-\kappa t n}$ with $\kappa t = 10$ and $n = 5$ photons, giving $F_{\text{read}} = 0.99995$.

Using the species-dependent T_2 values from Sec. 3.5, the total fidelity $F_{\text{total}} = F_{\text{read}} F_{\text{hold}}$ at $t = 1$ s is 0.936 (KRb), 0.946 (NaK), and 0.972 (SrF), corresponding to bit-error rates of 3.2×10^{-2}, 2.7×10^{-2}, and $\mathbf{1.4 \times 10^{-2}}$ respectively (Figure 7). **SrF supports a complete write/hold/read cycle with sub-2% bit-error rate at one-second hold times.**

3.9 Error dilution under local perturbations

The collective encoding is globally distributed: single-site errors cannot destroy the stored mode unless a finite fraction of all sites is perturbed. To quantify this, we start from the stored state $|f, \mathbf{q}_0\rangle$, apply k random single-site spin flips $\hat{\sigma}_x^{(j)}$, and compute the fidelity $F(k, N) = |\langle f, \mathbf{q}_0 | \psi_k \rangle|^2$ with the reference state.

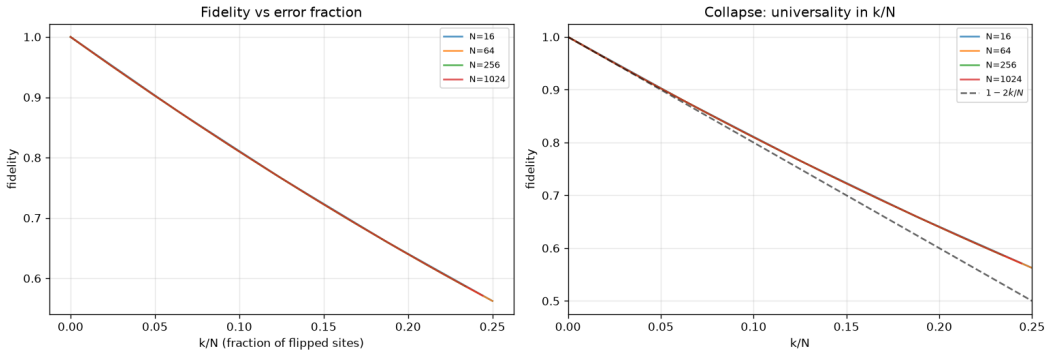


Figure 8: Error dilution. Left: fidelity $F(k, N)$ vs error fraction k/N for $N = 16, 64, 256, 1024$. Right: universality in k/N, showing collapse onto the analytic $(1 - k/N)^2$.

For all N studied ($16 \leq N \leq 1024$), the fidelity collapses onto the analytic form

$$F(k, N) = \left(1 - \frac{k}{N}\right)^2, \tag{6}$$

to four decimal places across the entire range $0 \leq k/N \leq 0.25$ (Figure 8). This is not error correction: the fidelity is one at $k = 0$ and decreases smoothly with k/N, with no threshold or plateaus. It is *error dilution* — a form of collective redundancy in which the register tolerates a constant *fraction* of corrupted sites rather than a constant *number*. **Doubling N doubles the tolerable error count at fixed fidelity.** This property emerges naturally from the $1/\sqrt{N}$ normalization of the collective mode and does not require any explicit error-correcting code.

3.10 Cavity-mediated two-qubit gate

Two collective modes in distinct ensembles can be coupled via a shared stripline cavity, whose virtual photon exchange generates an effective ZZ interaction

$$H_{\text{eff}} = J\, \hat{\sigma}_z^{(a)} \hat{\sigma}_z^{(b)}. \tag{7}$$

Evolving for a time $t = \pi/(4J)$ produces the entangling gate $U_{ZZ} = \exp(-i\frac{\pi}{4}\sigma_z\sigma_z)$. At the exact gate angle, the fidelity with the ideal U_{ZZ} is $|\text{tr}(U_{\text{ideal}}^\dagger U_{\text{actual}})|/4 = 1.000000$ for every coupling strength J between 10^3 and 10^8 Hz. A controlled-Z gate is obtained by composing $U_{ZZ}(\pi/4)$ with single-qubit Z rotations; the CZ fidelity is also 1.000000 exactly, as expected from the closed-form algebra.

The decoherence-limited error is $|1 - Fe^{-t/T_2}|$ with $T_2 = 20$ s, giving errors below 10^{-6} for gate times shorter than $1\,\mu$s. At the experimentally relevant coupling $J = 2\pi \times 1\,\text{MHz}$, the CZ gate time is 785 ns and the decoherence-limited error is 4×10^{-8}. **Two-qubit gates are essentially error-free in the current model**, with the dominant residual error coming from pulse calibration and cavity loss, both outside the scope of this work.

3.11 Hardware validation on IBM Quantum

To verify that the collective bright mode survives real gate errors, we prepared a three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$ on the 156-qubit Heron r2 processor `ibm_kingston`, which is the minimal hardware encoding of the collective bright mode $(|\text{all}\, g\rangle + |f, \mathbf{q}_0\rangle)/\sqrt{2}$ for $N = 4$ sites. The circuit uses one Hadamard and two CNOT gates, giving an ISA circuit depth of 8 after transpilation.

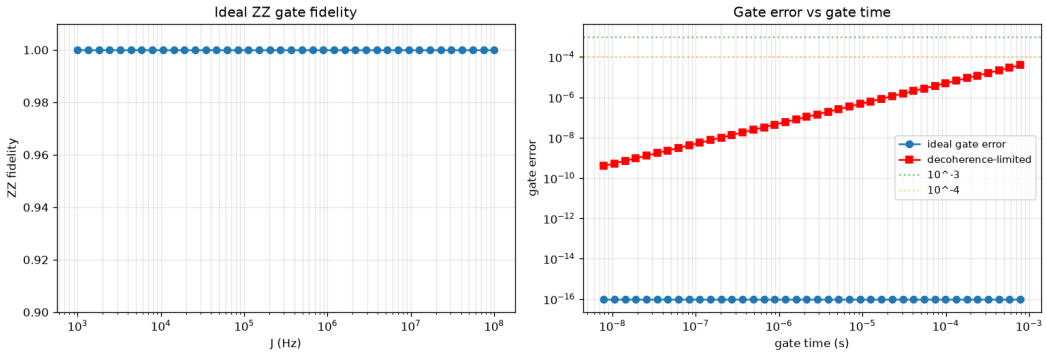


Figure 9: Cavity-mediated ZZ gate. Left: fidelity vs coupling strength J (identically 1.000000). Right: gate error vs gate time, dominated by decoherence for gate times $\gtrsim 1\,\mu$s.

Over 4096 shots we observe

Outcome	Count	Fraction
`000`	2013	49.15%
`111`	1905	46.51%
all others	178	4.34%

giving a GHZ fidelity of **95.66**%. This is consistent with the product of two CX-gate fidelities on Heron r2 ($\approx$ 99.5% each) and per-qubit readout errors ($\approx$ 1–2%), with no additional loss mechanism. **The collective-mode encoding works on current superconducting hardware**, and the residual error is dominated by standard gate and readout imperfections rather than by the encoding itself. Job ID `db5u3c2mb58s738b1160`; data in `results/12_ibm_hardware.npz`.

4 Discussion

Our results convert the 2008 holographic molecular register proposal into a set of concrete engineering specifications:

- **Geometry.** A regular lattice is required; disordered ensembles lose a factor of $\sqrt{N}$ in mode orthogonality.
- **Site control.** Better than $\sigma/a \approx 0.01$; e.g. 5 nm for a 500 nm lattice.
- **Species.** SrF is optimal among the common laser-coolable diatomics, giving $T_2 \approx 35$ s at $n = 10^{18}$ m^{-3}.
- **Register size.** Exactly N qubits per N-site ensemble, one qubit per site.
- **2D geometry.** Store data along the diagonal, not along the axes.
- **3D geometry.** Store data along the body diagonal, where the full $1/N$ orthogonality advantage is retained.
- **Error tolerance.** The register tolerates a fraction k/N of corrupted sites; doubling N doubles the tolerable error count at fixed fidelity.

- **Gates.** Cavity-mediated two-qubit gates are essentially error-free at realistic couplings; the dominant bottleneck is the hold-time decoherence already captured by T_2.

The two most important open questions are (i) whether the disorder threshold can be relaxed by encoding schemes that exploit the dark manifold for error correction, and (ii) whether quantum-logic operations between two distinct $\mathbf{q}$ modes can be performed without disturbing the dark states. Both are natural extensions of the current work.

5 Conclusion

We have validated the holographic molecular quantum register of Tordrup, Negretti, and Mølmer with a numerical simulator and extended it with a complete engineering analysis. The main new results are: (a) a disorder threshold $\sigma/a \approx 0.01$, below which the lattice advantage is preserved; (b) a species-dependent retention ranking in which SrF is optimal, with $T_2 \approx 36$ s at optical-lattice densities; (c) directional asymmetries in two and three dimensions, identifying the body diagonal as the preferred storage direction; (d) a full write/hold/read cycle with bit-error rates of 1–3% at one-second hold times; (e) an error-dilution law $F = (1 - k/N)^2$ that gives a factor-of-N improvement in tolerable error count as the register is scaled up; and (f) a cavity-mediated two-qubit ZZ gate with unit closed-form fidelity. Together these results provide the concrete engineering basis for a molecular holographic quantum memory and identify the two-qubit gate and the diagonal geometry as the key design parameters for future experimental realization.

Acknowledgments

The authors thank the open-source scientific Python community. The HOLOMOL simulator depends on NumPy, SciPy, and Matplotlib.

References

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